

Maksim Silchenko

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Education

National University of Singapore (NUS), Singapore

MSc in NUS School of Computing, specializing in Statistics

Jul '26 – Jan '28

SPAR Research Fellowship - Agent Attribution

Researcher · Mentor: Dr Yiming Li, Nanyang Technological University (NTU)

Sep '26 – Dec '26

- Recasting agent attribution as Bayesian inverse planning: posterior inference over candidate configurations (model, scaffold, prompt, memory), scored by the likelihood each assigns to the observed actions, with open-set rejection.
- Building a unified testbed (open-set, EER/DET, cross-environment) against published behavioural-fingerprinting classifiers (>95% closed-set), then intervention studies and activation probes to isolate which component carries the signal, toward a paper.

NUS MBA Research Project - LLM-Based Admissions Screening

Research Assistant · Advisor: Dr Hong Ming Tan, Dept of Analytics & Operations

Sep '26 – Dec '26

- Benchmarking frontier LLMs for MBA admissions screening vs human ratings and alternative benchmarks (repeated trials, CIs, statistical tests, noise).
- Training linear probes on the models' hidden states to find various biases hidden from the written rationale, then removing them inside the model; long-term, validating scores against students' improvement during the MBA and developing a unified protocol for similar AI screening tools.

HONOR-NUS Research Project - Recommender Systems

Research Assistant · Advisor: Prof. James Pang, NUS Business Analytics Centre

Aug '26 – Dec '26

- Research with HONOR on its millions-user phone feed: generative ranking over long user histories (HSTU, MTGR, HPGR) and multi-task ranking that fixes rare conversion targets (PLE vs ESM, AITM) with UniMixer for GPU optimisation and parameter scaling.
- Designing the filter-bubble study: if a less repetitive lock-screen feed raises retention, user satisfaction and ultimately prevents information bubble.

Programs & Bootcamps

• **TARA Singapore, ARENA** - Transformers & RL, mech interp, model evals with inspect, LLM agents, datasets generation.

Sep '26 – Dec '26

• **ML4Good, Hong Kong** (fully sponsored) – mech interp, alignment, RLHF, agentic AI, evals, risk modelling, forecasting.

Oct '26

• **BlueDot Impact** 'Technical AI Safety' Program (cohort-based, application-only) - Alignment & RLHF, mech interp, evals & red-teaming.

Aug '26 – Sep '26

• **University of Oxford Bootcamp** 'ML Representation Learning & Gen AI' - Bayesian ML, geometric DL, multimodal foundation models.

Jul '26

• **Beijing Univ. of Posts & Telecoms (BUPT) Bootcamp** 'Agentic AI' - Multi-agents, retrieval, LLM-as-a-judge, guardrails & escalation.

Jun '26 – Jul '26

Experience

Tazapay, Singapore

AI & Analytics Intern

Jun '26 – Sep '26

- Building an agentic automation harness in n8n, using LangGraph to orchestrate Claude API, AWS S3, and internal tools into a closed-loop monthly data audit.
- Developing and evaluating a company-wide finance chatbot (LangChain retrieval, LangSmith tracing), applying RLHF to improve responses from user feedback, alongside data engineering on ClickHouse and Tableau.
- Analysing new client cohorts with OLS, difference-in-differences, and A/B tests with guardrail metrics to measure performance against existing segments.

Orbuc Research, London, UK

Quantitative Researcher Intern

Feb '26 – Apr '26

- Developed a time-series anomaly-detection model that classifies latent states with Hidden Markov Models to flag unstable, low-predictability conditions, validated across 5 out-of-sample windows where it held up in 4 of 5.
- Built an LLM agent that automatically classifies events from live news streams, using retrieval-augmented generation (RAG) to ground each call, a probabilistic layer to aggregate outputs into a single risk score, and SQL (SQLite) storage to persist every event and classification [[Agent Prototype](#)].
- Engineered the stats analysis behind the firm's March '26 research report: found 9 historical anomaly periods and 4 significant shifts in correlation structure.

The TCM Group, London, UK

Data & Analytics Consultant Intern (Bayes Business School Capstone) · Top of the Cohort

Jan '26 – Apr '26

- Designed a causal-inference and hypothesis-testing framework (RCT-style wait-list control, distribution-aware paired tests reporting effect sizes with CIs against a meaningful-change threshold) to measure training outcomes, adopted by TCM as the standard for new programme evaluation.
- Engineered a serverless, rubric-based LLM scoring engine (TypeScript, Vercel, OpenAI, Zod-validated with automatic retry) that grades open-ended responses 0 to 10 at 0.90 Spearman with human raters, halving scoring error via a cross-validated bias calibration [[LLM Prototype](#)].

Oxford Comma Advisory, London, UK

Data Analyst Intern

Oct '25 – Jan '26

- Cut university-shortlisting time from 2 hours to 3 minutes per student by achieving a 73% consultant approval rate on top-5 recommendations through a two-tower neural network trained on sparse implicit feedback with confidence-weighted negative sampling.
- Found organic leads convert about 60% faster than paid channels and that early consultations carry a 2.3x hazard ratio, using Kaplan-Meier curves and a Cox proportional-hazards model on 800+ inquiries (non-converters right-censored).

Awards & Competitions

Google Cloud Rapid Agent Hackathon ('26): [1st Place](#) (Gold, \$5k) of 15k teams for AutoSRE, an autonomous on-call engineer [[Devpost](#)].

AKBC 2026 Shared Task @ EMNLP ('26), featured: [1st Place \(Gold\)](#), 0.7060 macro F1 on knowledge base construction from language models [[Page](#)].

CAR-Bench Challenge @ IJCAI-ECAI '26: [2nd place](#), Track 1 (Open) - reliability of tool-using LLM agents; technical report published [[Report](#)].

UbiComp '26, Trustworthy Agentic AI: [1st place \(Gold\)](#), paper accepted to the ACM UbiComp/ISWC '26 Companion proceedings (TCSAUC Workshop). [[Page](#)].

WMT26 MTEval Challenge Sets (EMNLP '26): QEBreak challenge set accepted and run over all metric submissions for presentation @WMT26, EMNLP;

BAREC Shared Task ('26, ArabicNLP @ EMNLP '26): [1st places](#), Strict and Open tracks; system paper "Ensembling at the Noise Floor" [[Page](#)].

BlueDot AI-Safety Grant Recipient ('26): Bayesian framework with calibrated uncertainty for whether LLM chain-of-thought is faithful [[Page](#)].

RecSys in HR Challenge (WorkRB, @ ACM RecSys '26): [1st Place \(Gold\)](#); invited to present at the workshop, Minneapolis, Sep '26.

ChildSafeAds 2026 Shared Task NLLP @ EMNLP ('26): system paper accepted, on advertising-law compliance in child-facing YouTube videos

Latent Probe Challenges ('26, Codabench): [1st place](#), Toxicity Detection and Mental Health Sentiment Classification, probes on Gemma 2 2B hidden states.

National Mathematics Olympiad (MIPT): [Winner](#), 0.015% nationally (Russia).

Jane Street 'Puzzle' Competition (Feb, Jun, Jul '26): [correct solutions found](#).

AlphaNova Quant Competition 5 ('26): ranked [9th of 873](#); [3 signals accepted](#) into the platform's uncorrelated quality set.

BlueDot Tech AI Safety Puzzle #1 ('26): [correct solution found](#), reverse-engineered a feature the network hid from linear probes [[Page](#)].

Relevant Projects

- Olist Brazilian Marketplace: Causal Inference & Bayesian Modelling** *(Independent Project)* [\[Project Page\]](#) [\[GitHub\]](#)

'26
- Reframed a flat marketplace A/B test on a 97k-order panel as a causal-identification problem, using a hierarchical Bayesian DiD in PyMC (DuckDB pipeline, NetworkX causal DAG, falsification tests) that flipped a naive -2pp read into a DiD-identified +1.5pp and a costed -R\$452K no-launch call.
- A/B-Test Experimentation Guardrail: SRM Detection & Causal Inference** *(Independent Project)* [\[Project Page\]](#) [\[GitHub\]](#)

'26
- Built a command-line experimentation guardrail that flags unsafe A/B tests via a chi-square SRM check (catching a 61/39 allocation a t-test misses), then de-confounds with propensity-score matching and Rosenbaum sensitivity bounds, validated on Criteo's 13.98M-row Uplift data.

Skills

Programming & DBs: Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, statsmodels, SciPy, XGBoost, PyMC), SQL, DuckDB, R, C++, TypeScript.

Machine Learning & AI: Supervised & Unsupervised Learning, GB, NNs (CNNs, RNNs, LSTMs, Transformers, Two-Tower), Reinforcement Learning (Q-learning, PPO, Actor-Critic), Rec Systems & Embeddings, LLMs, Prompt Engineering, RAG, Agentic AI (LangChain, LlamaIndex, tool-use).

Statistical Methods: Causal Inference, A/B Testing, Survival Analysis, Bayesian Statistics, Time-Series Analysis, Probability Calibration, RCTs.

Visualisation, Cloud & MLOps: Tableau, Power BI, matplotlib, Plotly; GCP, AWS, Docker; MLOps (CI/CD, GitHub Actions); Git, Jupyter, LaTeX.

Spoken Languages: English (Fluent), Russian (Native), Ukrainian (Native), Belarusian (Fluent), Spanish (Professional Working).

Latest Blog ‘What the Leaderboards Are Actually Measuring’ - A critique of the public evals metrics for frontier AI models (Fable 5.1 & Astra) [\[Page\]](#)